

- 9 The Hood Island giant tortoise (*Chelonoidis hoodensis*) is only found on one of the Galapagos Islands.

The photograph shows two Hood Island giant tortoises fighting over territory.



(Source: © Minden Pictures / Alamy Stock Photo)

- (a) Which of the following is a term that would be used to describe the fact that this species of giant tortoise is found only on the Galapagos Islands?

(1)

- ☐ A diversity
- ☐ B endemic
- ☐ C isolation
- ☐ D linkage



(b) The Hood Island giant tortoise has adaptations to its environment.

Two features are:

- long neck
- shell that arches above the neck.

When there is a dispute over territory, the tortoises use these features to make themselves look as large as possible.

(i) Complete the table to show the type of adaptations shown by this tortoise.

(3)

Feature	Type of adaptation
long neck	
shell that arches above neck	
making themselves look as large as possible	

(ii) Suggest one selection pressure that results in the development of one of these features.

(1)

.....

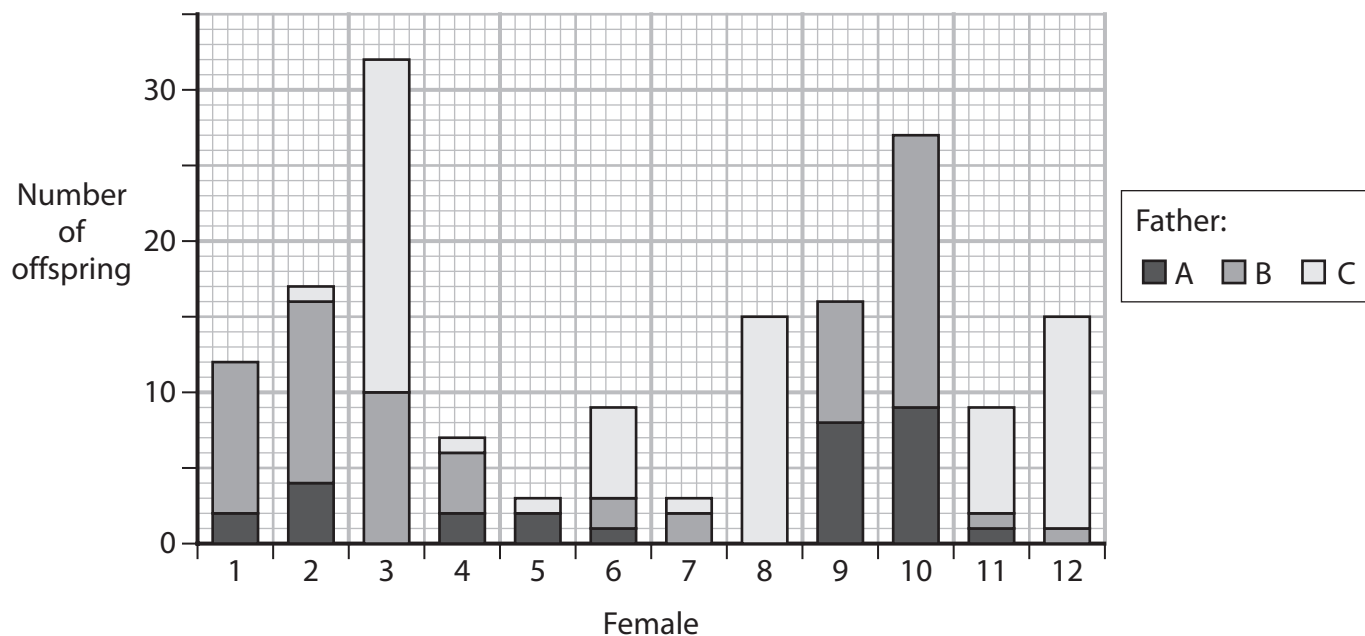
.....

.....

(c) In 1963, there were only three males and 12 females of *C. hoodensis* left in the wild.

These 15 tortoises were used in a breeding programme.

The diagram shows the number of offspring from each female tortoise and the fathers of these offspring.



- (i) Calculate the percentage of female 10's offspring that were fathered by male B.

(1)

Answer%

- (ii) The wild population of *C. hoodensis* is now 1800 individuals.

Calculate the percentage increase in the wild population of *C. hoodensis* tortoises.

Give your answer to **one** decimal place.

(2)

Answer%



- (iii) A zoo tested its captive-bred giant tortoise and determined that it was a *C. hoodensis* tortoise.

Explain how the zoo determined that its giant tortoise was a *C. hoodensis* tortoise.

(3)

- (d) Scientists thought that a recessive allele was changing in frequency in the tortoise population.

Explain how the change in frequency of this allele could be determined.

(2)

(Total for Question 9 = 13 marks)

TOTAL FOR PAPER = 80 MARKS

